CardioPredict — Project Setup, venv & pip

Practical Coding Notes | YUVA Internship

1. Project Setup

Project: CardioPredict — a Machine Learning project for heart disease prediction.

What we did: Created the CardioPredict GitHub repository and project folder.

Practice file: sample.py

2. Create and Activate venv

We created a project-specific Python virtual environment and activated it before installing packages.

The important idea: the project uses its own isolated Python environment.

Basic venv Commands

```
python -m venv venv

# Windows PowerShell
.\venv\Scripts\Activate.ps1

# Deactivate
deactivate
```

3. What venv Solves

Different projects can use different package versions without interfering with each other.

The venv folder is not pushed to GitHub. The environment can be recreated later from a dependency list.

4. pip

pip is the command-line tool we used to install Python packages.

We installed Pandas inside the activated CardioPredict environment.

pip Commands Practiced

```
pip install pandas

pip show pandas

pip list

pip --version
```

5. Package and Dependency Learning

Pandas can require other packages. During installation, NumPy was also installed as a dependency.

We also learned that installing a package in one environment does not automatically make it available in another environment.

6. Debugging We Practiced

We encountered ModuleNotFoundError: No module named 'pandas'.

The problem was related to running the code with a Python environment where Pandas was not installed.

The practical lesson: check that the correct venv is activated and that the package is installed in that environment.

7. Current Project Structure

At this stage, the project contains the CardioPredict workspace, the virtual environment, the sample Python file, and the practice patient CSV dataset.

The venv itself remains local and is not part of the GitHub source code.

8. Quick Revision

venv → creates an isolated Python environment.

activate → starts using that environment.

pip install → installs a package.

pip list → shows installed packages.

deactivate → leaves the environment.